Shift-Aligned Reduced-Order Models and Weak-Form PDE Discovery for Interacting-Agent Densities from Noisy Trajectories

Honors Thesis Defense

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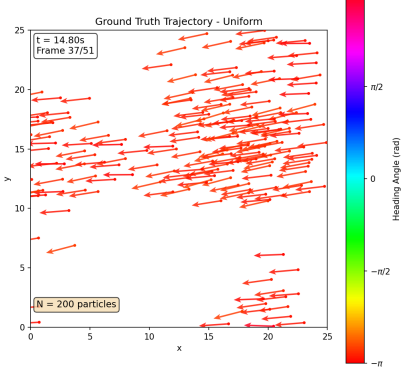
Advisor: Prof. Björn Sandstede · Second Reader: Prof. Ian Wong · April 2026

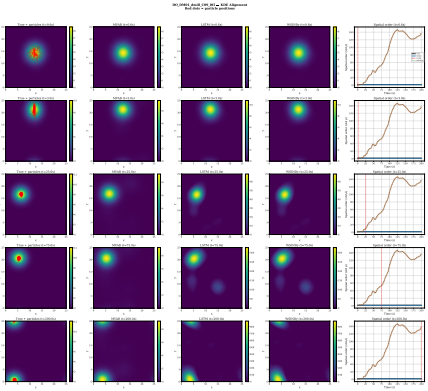


particles

N=300 self-propelled agents on $\mathbb{T}^2$

KDE

h=5 cells



coarse fields: ρ, $\mathbf{p}$, Φ

$$\rho(\mathbf{x},t) = \sum_{i=1}^{N} K_h(\mathbf{x}-\mathbf{x}_i(t)), \quad \mathbf{p}(\mathbf{x},t) = \sum_{i=1}^{N} K_h(\mathbf{x}-\mathbf{x}_i(t))\,\mathbf{v}_i(t), \quad \Phi = W * \rho$$

Two Headline Results

Hypothesis: the aligned latent dynamics are near-linear.

Confirmed: MVAR matches LSTM accuracy with **116**× fewer parameters — model capacity is not the bottleneck.

$R^2_{\mathrm{wf},\rho}$>0.99 in 8 of 9 regimes does not guarantee the PDE can be rolled out autonomously.

Identification ≠ closure: the dominant-balance library sees no saturation signal, so autonomous rollout diverges.

Strong weak-form fit can coexist with failed autonomous rollout.

Two Pipelines, Two Questions

$$\partial_t \rho + \nabla \cdot \mathbf{j}_{\boldsymbol{\theta}}[\rho, \mathbf{p}, \Phi] = 0 \quad \longleftarrow \textbf{shared closure target}$$

Part I: EF-ROM

- Shift-aligned POD + latent MVAR or LSTM
- Question: **forecastability** — how well can we predict collective motion?
- Score: closed-loop R^2_{fc}

$$\mathbf{y}_k = U_m^\top(\tilde{\mathbf{x}}_k - \boldsymbol{\mu}), \quad \hat{\mathbf{y}}_k = A_0 + \sum_{j=1}^{w} A_j \mathbf{y}_{k-j}$$

$$z_{k+1} = \mathcal{A}_{\mathrm{comp}}\, z_k + b_{\mathrm{comp}}, \quad \rho(\mathcal{A}_{\mathrm{comp}}) < 1$$

Part II: WSINDy

- Weak-form sparse regression on field data
- Question: **identifiability** — what PDE governs the density?
- Score: weak-form $R^2_{\mathrm{wf},\rho}$

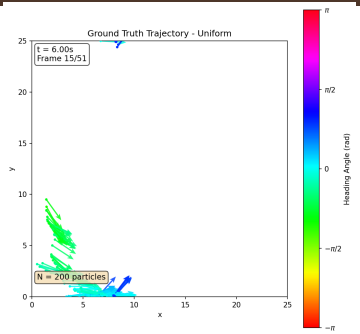
$$b_k = -\int_{Q_k} u\, \partial_t \psi_k, \quad \mathbf{b} = \mathbf{G}\mathbf{w}$$

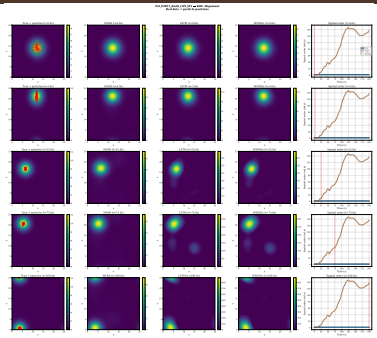
$$\mathbf{w}^* = \mathrm{MSTLS}(\mathbf{G}, \mathbf{b}, \lambda), \quad R^2_{\mathrm{wf}} = 1 - \frac{\|\mathbf{b} - \mathbf{G}\mathbf{w}^*\|^2}{\|\mathbf{b}\|^2}$$

Rollout is a **closure stress test**, not the main identifiability score.

$\Rightarrow \quad R^2_{\mathrm{wf},\rho} > 0.99$ *and* rollout diverges — these are compatible.

From Particles to the Shared Field Space





Representative regime (blackhole): 300 agents, strong Morse attraction (C_a=20)

Coarse-grained fields: density ρ, polarisation $\mathbf{p}$, Morse field Φ

$$\rho = \sum_i K_h(x - x_i), \qquad \mathbf{p} = \sum_i K_h(x - x_i)\mathbf{v}_i, \qquad \Phi = W * \rho$$

Same KDE pipeline applied to all 9 regimes $\times$ 4 IC families.

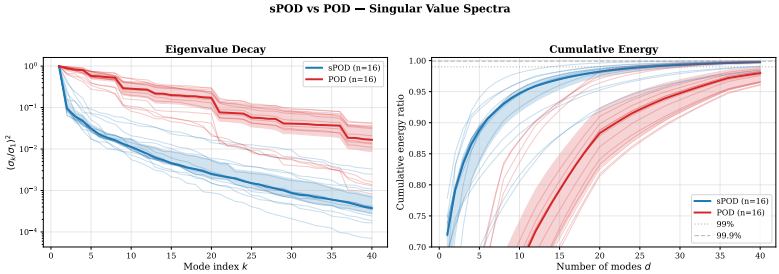
Shared Benchmark Design

Element	Details
Regimes	Gas, Blackhole, Supernova, Crystal (CS + VS variants), Pure Vicsek
IC families	Gaussian, Ring, Two clusters, Uniform
Train/test split	340 training runs + 4 held-out test runs per regime
Data source	Same KDE coarse-graining, bandwidth h=5 cells
ROM branch	Shift-aligned POD (n_δ) + MVAR or LSTM in POD space
WSINDy branch	Weak-form sparse regression in physical field space
Shared evaluation	Same test ICs, same rollout split, both branches

9 regimes **4 IC families** **shared KDE fields** **same train/test split**

N=300, 64×64 grid, h=5 cells, Δt_{ROM}=0.12 s · VS: force changes speed · CS: heading only

Shift Alignment Halves the Effective Rank



$$\tilde{\rho} = \sqrt{\rho + \varepsilon},\ \varepsilon = 10^{-10} \quad \text{(variance-stabilising transform before SVD)}; \quad \widetilde{X} \in \mathbb{R}^{d \times T}$$

$$\boldsymbol{\delta}_k^* = \arg\max_{\boldsymbol{\delta} \in G} C_k(\boldsymbol{\delta}), \qquad C_k(\boldsymbol{\delta}) = \langle \rho_{\mathrm{ref}},\, \mathcal{T}_{\boldsymbol{\delta}} \rho_k \rangle$$

$$\mathcal{E}(m) = \frac{\sum_{i=1}^{m} \sigma_i^2}{\sum_{i=1}^{M} \sigma_i^2}, \qquad m{=}19:\ \mathcal{E}(19) = 0.976 \qquad \left(r_{\mathrm{eff}}^{\mathrm{raw}} \approx 38,\ r_{\mathrm{eff}}^{\mathrm{aligned}} = 19\right) \ \mathbf{y} = U_m^\top(\tilde{\mathbf{x}} - \boldsymbol{\mu})$$

$$\textbf{[restrict]} \qquad \hat{\tilde{\mathbf{x}}} = U_m \mathbf{y} + \boldsymbol{\mu} \ \ \textbf{[lift]} \qquad \hat{\rho} = [\max(0, \hat{\tilde{\mathbf{x}}})]^2 - \varepsilon \ \ \textbf{[invert]}$$

Efficiency Summary: MVAR vs LSTM

$$p_{\mathrm{MVAR}} = r + wr^2 = 19 + 5{\cdot}19^2 = 1{,}824 \ll p_{\mathrm{LSTM}} = 211{,}091 \quad (116\times \text{ more})$$

$$\hat{B} = \left(X^\top X + \alpha\Gamma\right)^{-1} X^\top Y, \quad \alpha = 10^{-4}; \qquad z_{k+1} = \mathcal{A}_{\mathrm{comp}}\, z_k + b_{\mathrm{comp}}, \quad \rho(\mathcal{A}_{\mathrm{comp}}) \text{ as stability diagnostic}$$

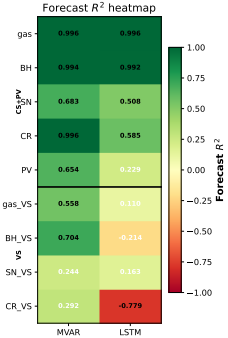
Metric	**MVAR**	**LSTM**
Trainable parameters	1,824	211,091
Training time	<1 s (closed-form)	$\sim$30 min (GPU Adam)
Median test R^2 (CS+PV)	0.995	0.585
Median test R^2 (VS)	0.425	-0.052

MVAR linear-in-latent hypothesis passes: closed-form training, companion-spectrum stability diagnostics, 116× fewer parameters.

ROM Forecast Quality Across Regimes

$$R^2_{\mathrm{fc}} = 1 - \|\rho^* - \hat{\rho}\|_F^2 \,/\, \|\rho^* - \bar{\rho}\|_F^2 \quad \text{(closed-loop rollout, held-out test ICs)}$$

$z_{k+1} = \mathcal{A}_{\mathrm{comp}} z_k + b_{\mathrm{comp}}$; companion spectrum gives a linear stability diagnostic for MVAR



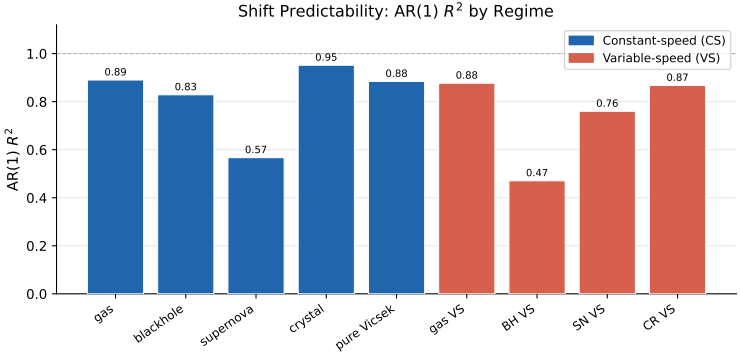
Median CS+PV: R^2=0.995; MVAR dominates the main benchmark; parity only in the easiest regimes. **VS block:** ceiling set by shift AR: $R^2_{\mathrm{AR}}(\mathrm{BH\,VS})$=0.47

Shift Predictability: A Structural Bottleneck

$\delta(t) = a_0 + \sum_{k=1}^{p} a_k\, \delta(t-k) + \varepsilon_t, \quad p{=}5$ — $R^2_{\mathrm{AR}} \downarrow$ tracks weaker shift predictability and, in practice, lower R^2_{fc}

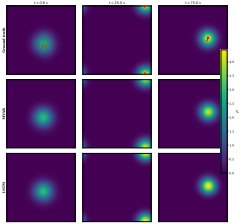
Gas CS: $R^2_{\mathrm{AR}} = 0.89 \rightarrow R^2_{\mathrm{fc}} = 0.995$; **BH VS:** $R^2_{\mathrm{AR}} = 0.47 \rightarrow R^2_{\mathrm{fc}} = 0.43$

Interpretation: structural bottleneck of the single-reference alignment representation, not a formal upper-bound theorem.



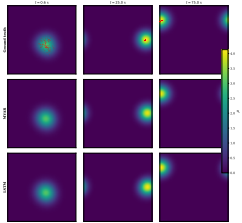
When the ROM Works vs. When It Fails

Gas (CS) — shift is predictable



$R^2_{\mathrm{AR}} = 0.89 \Rightarrow R^2_{\mathrm{fc}} = 0.995$: shape *and* location match

Blackhole VS — shift is weakly predictable



$R^2_{\mathrm{AR}} = 0.47 \Rightarrow R^2_{\mathrm{fc}} = 0.43$: right shape, wrong location

$\rho^*(x,t) \approx \rho_{\mathrm{ROM}}(x - \boldsymbol{\delta}_{\mathrm{err}}(t), t)$ — shape tracked, bulk translation missed

In tests, a two-stage corrector (predict $\boldsymbol{\delta}$ separately, apply to ROM output) did not help, suggesting the shift process is not predictable enough at this scale.

WSINDy: Move Derivatives Off the Noisy Data

Target PDE $u_t = \sum_{m=1}^{M} w_m \mathcal{L}_m[f_m(u)]$

Test functions: separable compactly supported bumps

$$\psi_k(x,y,t) = \phi\left(\frac{x-x_k^0}{\ell_x}\right)\phi\left(\frac{y-y_k^0}{\ell_y}\right)\phi\left(\frac{t-t_k^0}{\ell_t}\right), \qquad \phi(s) = \left(1-s^2\right)_+^q$$

Weak form on Q_k:

$$b_k := -\int_{Q_k} u\,\partial_t\psi_k = \sum_m w_m \int_{Q_k} \mathcal{L}_m^*[\psi_k]\, f_m(u) =: \sum_m G_{km} w_m$$

System: $\mathbf{b} = \mathbf{G}\mathbf{w}$, solved by $\mathbf{w}^* = \mathrm{MSTLS}(\mathbf{G}, \mathbf{b}, \lambda)$.

- **Derivatives act on** ψ_k, not on the noisy field u — integration by parts (IBP):
  - temporal boundary terms vanish: ψ_k has compact support in t
  - spatial boundary terms vanish: domain is periodic
- **MSTLS:** keep term j if $\hat{\lambda} \le \Pi_j \le 1/\hat{\lambda}$, $\quad \Pi_j = |\tilde{w}_j|\,\|\tilde{G}_{:,j}\|_2/\|\mathbf{b}\|_2$.
- $R^2_{\mathrm{wf},\rho} = 1 - \|\mathbf{b}_\rho - \mathbf{G}_\rho \mathbf{w}_\rho^*\|^2/\|\mathbf{b}_\rho\|^2$.

Per-Regime Identification Verdict

$$R^2_{\mathrm{wf},\rho} = 1 - \|\mathbf{b}_\rho - \mathbf{G}_\rho \mathbf{w}^*_\rho\|^2 \,/\, \|\mathbf{b}_\rho\|^2$$

Regime	$R^2_{\mathrm{wf},\rho}$	$R^2_{\mathrm{wf},p}$	Closure	Verdict
gas (CS)	0.9998	0.69	partial	strong
blackhole (CS)	0.9995	0.79	partial	strong
supernova (CS)	0.9990	0.34	inadequate	moderate
crystal (CS)	0.9999	0.58	partial	strong
pure Vicsek	1.0000	0.82	adequate	strong
gas_VS	0.9990	0.74	partial	strong
blackhole_VS	0.9520	0.21	inadequate	partial
supernova_VS	0.9970	0.39	inadequate	moderate
crystal_VS	0.9999	0.86	inadequate	strong

- The continuity backbone is recovered in all nine regimes.
- The main deterioration is in the momentum equations, not in the density equation.
- Momentum deteriorates because $\mathbf{p} = \sum_i K_h(x - x_i)\mathbf{v}_i$ inherits velocity noise; the density is a smoother kernel sum and therefore a better weak-form target.
- Closure is reported separately because identification and rollout can decouple.

Strong-ID criteria: `div_p` coeff $\in [-1.01, -0.95]$; $R^2_{\mathrm{wf},\rho} > 0.99$; plausible signs; support stable across 6 test-function scales.

Pure Vicsek: WSINDy Validates against Kinetic Theory

Kinetic theory suggests:

$$\partial_t \rho + v_0 \nabla \cdot \mathbf{p} = 0$$

$(p_x)_t$ includes pressure, self-advection, viscous diffusion, and alignment relaxation; $W \equiv 0$ (no Morse).

WSINDy recovers:

$$\rho_t = -\nabla \cdot \mathbf{p}$$

$$(p_x)_t = \beta\, \partial_x \rho + \delta\, \nabla \cdot (p_x \mathbf{p})$$

$R^2_{\mathrm{wf},\rho} = 1.000$, $R^2_{\mathrm{wf},p} = 0.82$

Normalised balance: $\Pi_\delta = 1.000$, $\Pi_\beta = 0.963$

No spurious Morse term selected; diffusion and relaxation fall below threshold.

Correct leading-order structural recovery from data alone.

Gas Regime: Discovered PDE

Density equation

$$\rho_t = -\nabla\cdot \mathbf{p} - c\,\nabla\cdot(\rho\nabla\Phi)$$

Momentum equation

$$(p_x)_t = \beta\,\partial_x\rho + \nu\,\Delta p_x + \gamma\,\rho\,\partial_x\Phi + \delta\,\nabla\cdot(p_x\mathbf{p})$$

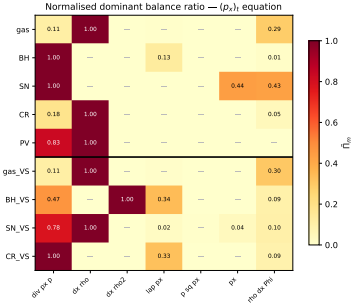
$$R^2_{\mathrm{wf},\rho} = 0.9998, \quad R^2_{\mathrm{wf},p} = 0.69$$

Term	Physics	$\Pi_m/\Pi_{\max}$
$\rho\,\partial_x\Phi$	Morse aggregation	1.000
$\partial_x\rho$	pressure	0.852
$\nabla\cdot(p_x\mathbf{p})$	self-advection	0.418
Δp_x	diffusion	0.128

$$\Pi_m = |w_m^*|\,\|\mathbf{G}_{:,m}\|_2/\|\mathbf{b}\|_2$$

Momentum Equation: Cross-Regime Term Selection

$$(p_\alpha)_t = \underbrace{\beta\,\partial_\alpha \rho}_{\text{pressure}} + \underbrace{\delta\,\nabla\cdot(p_\alpha \mathbf{p})}_{\text{self-advection}} + \underbrace{\gamma\,\rho\,\partial_\alpha \Phi}_{\text{Morse forcing}} + \underbrace{\nu\,\Delta p_\alpha}_{\text{diffusion}}, \quad \alpha \in \{x, y\}$$



$\partial_x \rho$: **pressure** | $\nabla\cdot(\rho_x \mathbf{p})$: **self-advection** | $\rho\,\partial_x \Phi$: **Morse forcing**

Self-advection and pressure are universal; Morse forcing discriminates attraction/repulsion regimes.

Pure Vicsek row: Morse column absent. · Attractive regimes: Morse column active.

Identification Success, Closure Failure

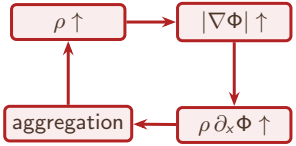
Identified law

$$\rho_t \approx -\nabla\cdot\mathbf{p} - c\,\nabla\cdot(\rho\nabla\Phi)$$

$R^2_{\mathrm{wf},\rho}$>0.999 on held-out data.

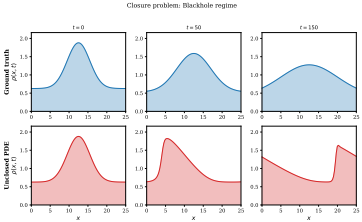
Missing: a saturation / congestion term. The saturation candidate $\Delta(\rho^2)$ is genuinely sub-dominant ($\Pi_\gamma<\hat{\lambda}$) in the training window, so weak-form fit cannot see it.

Unarrested feedback loop



No mechanism bounds ρ from above.

Synthetic closure schematic



top: truth · bottom: unclosed rollout

1D synthetic slice; 2D blackhole rollout is bounded but drifting

Strong identification can coexist with failed autonomous rollout.

What the Thesis Showed

1. **Scientific:** $R^2_{\mathrm{wf},\rho}>0.999 \not\Rightarrow R^2_{\mathrm{fc}}>0$ — identification and autonomous rollout decouple; rollout is a closure stress test.
2. **Pipeline:** $\tilde{\rho} \xrightarrow{U_m^\top} \mathbf{y} \xrightarrow{\text{MVAR}} \hat{\mathbf{y}} \xrightarrow{U_m} \hat{\tilde{\rho}}$ coupled to WSINDy on the same coarse-grained fields.
3. **EF-ROM:** MVAR reaches $R^2_{\mathrm{fc}}=0.995$ (CS+PV) with $p_{\mathrm{MVAR}}=1{,}824$ vs. LSTM $p=211{,}091$; VS ceiling set by $R^2_{\mathrm{AR}} \approx 0.47$.
4. **WSINDy:** $R^2_{\mathrm{wf},\rho}>0.99$ in 8/9 regimes; continuity backbone $\partial_t\rho+\nabla\cdot\mathbf{p}=0$ universal; missing saturation $\Rightarrow$ closure failure.

116× fewer params | $R^2_{\mathrm{fc}}=0.995$ | **8/9 regimes identified** | $R^2_{\mathrm{wf},\rho}>0.99$ | **closure $\neq$ identification**

Future Directions

Richer closure. Augment the density law with a porous-medium-type saturation that arrests Morse aggregation feedback $\rho \uparrow \Rightarrow |\nabla\Phi| \uparrow \Rightarrow$ blowup (congestion / excluded-volume interpretation; require $\gamma \geq 0$ for well-posedness):

$$\rho_t = -\nabla\cdot\mathbf{p} - c\,\nabla\cdot(\rho\nabla\Phi) - \gamma\,\Delta(\rho^2)$$

Geometry. Extend shift alignment to curved domains by replacing torus translations with the isometry group of the manifold $\mathcal{M}$,

$$\delta_k^* = \arg\max_{g\in\mathrm{Iso}(\mathcal{M})} \langle \rho_{\mathrm{ref}}, \mathcal{T}_g\rho_k\rangle_{L^2(\mathcal{M})}, \qquad \mathrm{Iso}(\mathbb{T}^2) = \mathbb{Z}_{n_y}\times\mathbb{Z}_{n_x}$$

Hybrid identification. Couple WSINDy support selection to closed-loop rollout error, retaining terms only when they improve both $R^2_{\mathrm{wf},\rho}$ *and* R^2_{fc}; this directly targets the identification-versus-closure gap.

Thank you.

Questions?

Advisor: Prof. Björn Sandstede • Second Reader: Prof. Ian Wong

Backup Slides

Additional material for discussion

Density Equation: Identified Coefficients Across All Regimes

Term	gas	BH	SN	CR	PV	gVS	BH_VS	SN_VS	CR_VS
div_p	-0.994	-0.986	-0.984	-1.001	-0.995	-0.987	-0.950	-0.978	-0.991
div_rho_gradPhi	-0.008	-2×10^{-4}	—	-2×10^{-4}	—	-0.006	-3×10^{-4}	—	-8×10^{-4}
lap_rho	—	$+0.007$	—	—	—	—	—	—	—
lap_rho2	—	-0.027	—	—	$+0.018$	—	—	$+0.008$	—
lap_p_sq	—	$+0.002$	—	—	-0.002	—	-3×10^{-5}	-4×10^{-4}	—

div_p ≈ -1 is the universal continuity backbone.

Momentum Equation: Identified Coefficients Across All Regimes

Term	gas	BH	SN	CR	PV	gVS	BH_VS	SN_VS	CR_VS
px	—	—	-0.219	—	—	-0.014	—	-0.077	—
p_sq_px	4×10^{-4}	8×10^{-5}	$+0.002$	$+0.001$	—	—	—	-0.008	-6×10^{-5}
dx_rho	-3.173	—	—	-2.179	-0.840	-4.379	—	-1.819	—
lap_px	-0.023	-0.048	—	—	—	-0.037	-0.117	-0.043	-0.162
rho_dx_Phi	$+0.916$	-0.005	$+0.214$	$+0.104$	—	$+1.331$	-0.033	-0.181	-0.043
div_px_p	-0.348	-0.369	-0.499	-0.399	-0.693	-0.486	-0.161	-1.419	-0.495
dx_rho2	—	—	—	—	—	—	-0.342	—	—

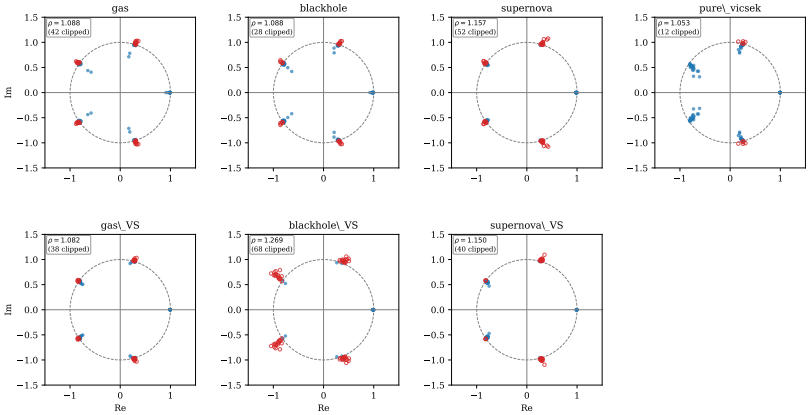
rho_dx_Phi sign: $+$ attraction, $-$ repulsion. div_px_p universally retained.

Full Regime Parameter Table

Label	Nickname	v_0	η	C_a	C_r	ℓ_a	ℓ_r	Behaviour
NDYN04	gas	3.0	1.5	0.5	0.2	1.5	0.5	disordered gas
NDYN04_VS	gas (VS)	3.0	1.5	0.5	0.2	1.5	0.5	variable-speed gas
NDYN05	blackhole	4.0	0.15	20.0	0.05	1.5	0.3	near-singular spike
NDYN05_VS	blackhole (VS)	4.0	0.15	20.0	0.05	1.5	0.3	variable-speed collapse
NDYN06	supernova	3.0	0.15	0.05	15.0	1.5	0.5	explosive repulsion
NDYN06_VS	supernova (VS)	3.0	0.15	0.05	15.0	1.5	0.5	variable-speed explosion
NDYN07	crystal	2.0	0.1	3.0	3.0	2.0	0.5	balanced equilibrium
NDYN07_VS	crystal (VS)	2.0	0.1	3.0	3.0	2.0	0.5	variable-speed equilibrium
NDYN08	pure Vicsek	3.0	0.3	—	—	—	—	alignment only

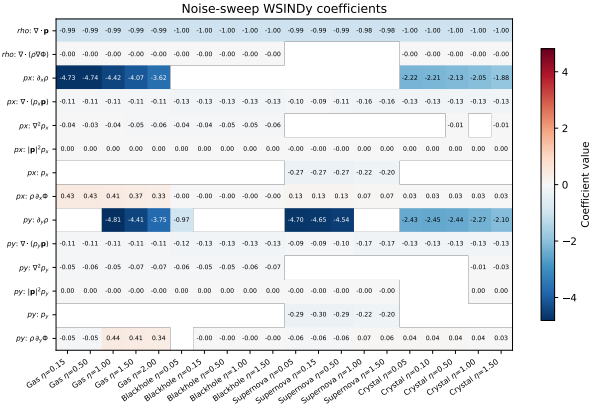
Shared: N=300, L=25, Δt=0.04 s, R=4.0, KDE 64×64, σ=5.0 grid cells, stride=3 (Δt_{ROM}=0.12 s), r=19 POD modes.

MVAR Companion Matrix Eigenvalue Spectra



Post-hoc spectral radius clamping stabilizes 2–3 modes but does not recover forecast accuracy — the loss is structural, not a stability artifact.

Noise Robustness (Appendix)



Completed noise-sweep WSINDy runs retain the same continuity backbone and a highly stable active set across the tested noise amplitudes.

MSTLS Algorithm: Sequential Thresholding

1. **Initialize:** OLS fit $\hat{\mathbf{w}} = \mathbf{G}^{\dagger}\mathbf{b}$
2. **Compute** dominant-balance ratios: $\Pi_m = \frac{|w_m| \, \|\mathbf{g}_m\|_2}{\|\mathbf{b}\|_2}$
3. **Threshold:** set $w_m{=}0$ if $\Pi_m < \alpha \max_j \Pi_j \quad (\alpha{=}0.1)$
4. **Re-fit** on active support: $\hat{\mathbf{w}}_{\mathcal{S}} = \mathbf{G}_{\mathcal{S}}^{\dagger}\mathbf{b}$
5. **Iterate** steps 2–4 until support stabilizes
6. **Cross-scale:** repeat for 6 support-scale triplets, retain intersection

Dominant-balance criterion (Callaham et al. 2021) normalizes by column norm, preventing large-scale terms from dominating.

From Agents to PDEs: The Closure Problem

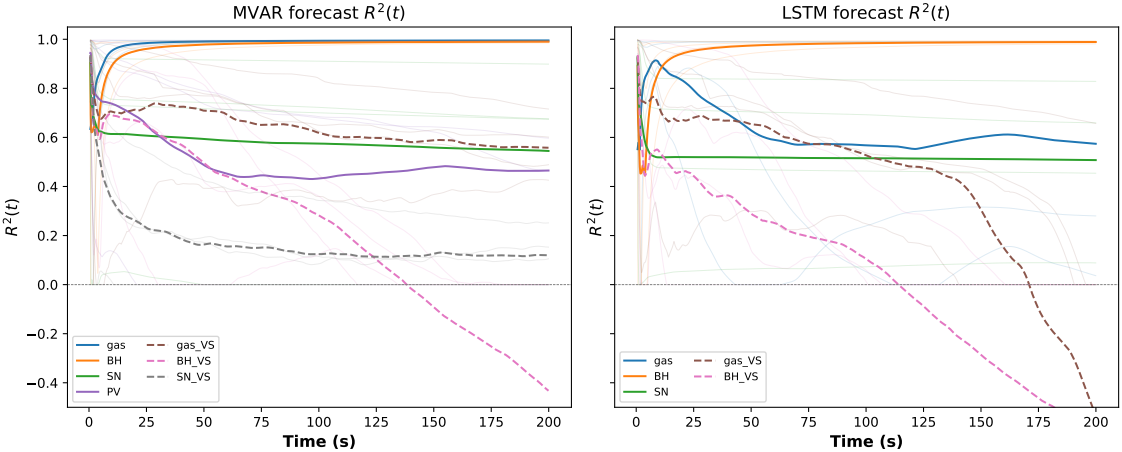
From microscopic trajectories the thesis observes $\mathbf{q} = (\rho, p_x, p_y, \Phi)$ with $\Phi = W * \rho$. The density must satisfy a **conservation law**:

$$\partial_t \rho + \nabla \cdot \mathbf{j} = 0.$$

The **closure problem**: identify an effective flux law $\mathbf{j}_{\boldsymbol{\theta}}[\rho, \mathbf{p}, \Phi]$ such that the above matches data.

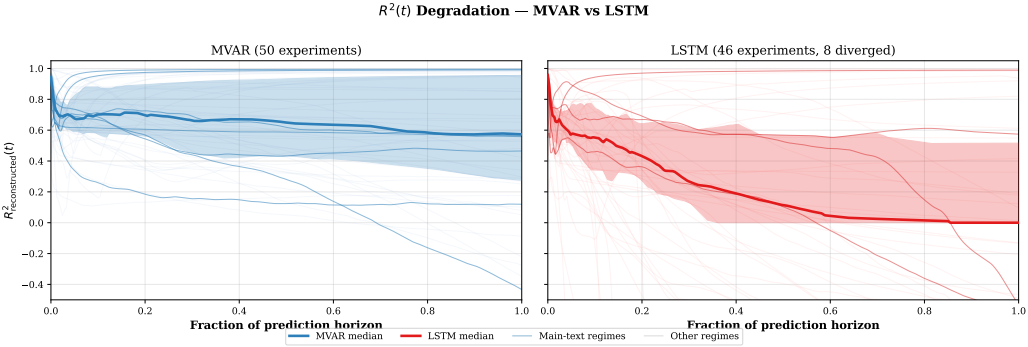
Scalar density alone does not close: alignment survives in $\mathbf{p}$; Morse interactions survive in Φ.

Forecast Degradation Over Time



R^2 Degradation Over Time: Full Experiment Suite

Faint traces = individual runs; bold = median; shading = IQR. Diverged LSTM runs ($R^2 < -2$) clipped at axis.



Blue (MVAR): median stays near 1 for CS, degrades for VS. **Red (LSTM):** higher variance across all groups; faster degradation on VS and blackhole.

Mass Preservation under Linear POD

Theorem (linear POD preserves mass)

If $\mathbf{x}_k = U_m \mathbf{y}_k + \boldsymbol{\mu}$ and $\mathbf{1}^\top U_m = \mathbf{0}$ (mean-centred basis), then

$$\mathbf{1}^\top \mathbf{x}_k = \mathbf{1}^\top \boldsymbol{\mu} = M_0 \quad \forall\, k,$$

so the lifted reconstruction has the same total mass as the training mean.

- In practice, the $\sqrt{\cdot}$ transform breaks linearity: reconstruction requires mass rescaling $\mathcal{S}_{M_0}$.
- Mass overshoot in any lifted snapshot is $\leq 0.3\%$ in all tested regimes.

Weak-Form Design Matrix Assembly

$$b_k = -\sum_{i,j,n} \partial_t \psi_k(x_i, y_j, t_n) \cdot u_{ij}^n \, \Delta x \, \Delta y \, \Delta t,$$

$$G_{k,m} = \sum_{i,j,n} \mathcal{L}_m^*[\psi_k])(x_i, y_j, t_n) \cdot f_m(u_{ij}^n) \, \Delta x \, \Delta y \, \Delta t.$$

- Separable test functions $\psi_k = \phi_x \phi_y \phi_t$ make convolutions efficient via FFT.
- The resulting system $\mathbf{b} = \mathbf{G}\mathbf{w}$ is tall-and-thin: many weak-integral constraints, few candidate library terms.
- Test-centre locations (x_k^0, y_k^0, t_k^0) are sampled uniformly inside the spatio-temporal domain.

Dominant-Balance Thresholding (MSTLS)

Term m is retained if its normalised contribution exceeds the threshold:

$$\Pi_m = \frac{|w_m| \, \|\mathbf{G}_{:,m}\|_2}{\|\mathbf{b}\|_2}, \qquad \hat{\lambda} \leq \Pi_m \leq \frac{1}{\hat{\lambda}}.$$

Steps:

1. OLS initialisation: $\hat{\mathbf{w}} = \mathbf{G}^{\dagger}\mathbf{b}$
2. Compute Π_m for all terms; threshold at $\hat{\lambda} \cdot \max_j \Pi_j$
3. Re-fit on active support; iterate until support stabilises
4. Repeat for 6 support-scale triplets; retain intersection

Synthetic Closure Model: the Missing Saturation

From WSINDy (identified): $R^2_{\mathrm{wf},\rho} > 0.99$

$$\rho_t = -\nabla\cdot\mathbf{p} - c\,\nabla\cdot(\rho\nabla\Phi)$$

For stable autonomous rollout (needed):

$$\rho_t = -\nabla\cdot\mathbf{p} - c\,\nabla\cdot(\rho\nabla\Phi) - \gamma\,\Delta(\rho^2)$$

The $-\gamma\,\Delta(\rho^2)$ term is a nonlinear diffusion that arrests the Morse aggregation feedback $\rho\uparrow \Rightarrow\ |\nabla\Phi|\uparrow \Rightarrow$ blowup.

- The saturation term is sub-dominant in the training window: $\Pi_\gamma < \hat{\lambda}$ (correctly thresholded away).
- Strong weak-form fit and failed rollout are therefore *compatible*.
- Richer library or kernel-based closure could capture saturation; left for future work.

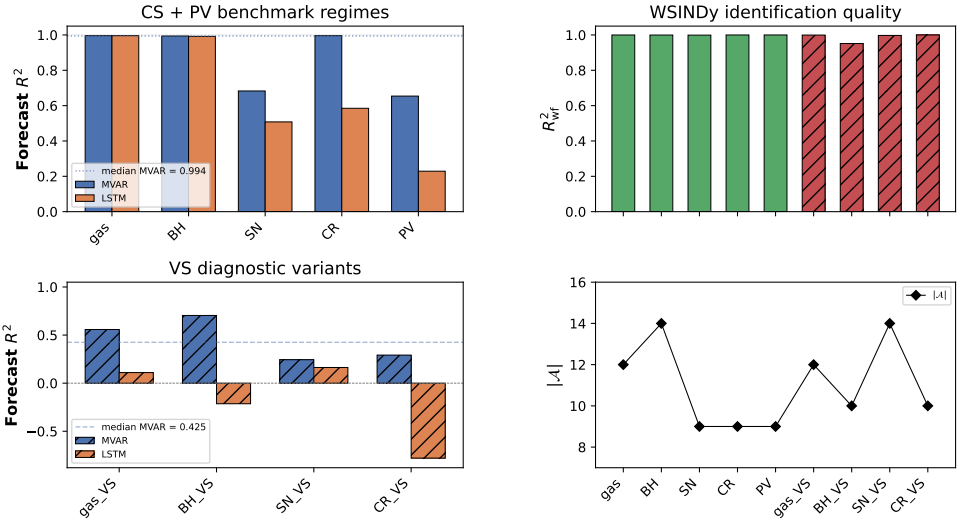
Identification Quality Is Measured in Weak Form

$$R^2_{\mathrm{wf}} = 1 - \frac{\|\mathbf{b} - \mathbf{G}\mathbf{w}^*\|_2^2}{\|\mathbf{b}\|_2^2}, \qquad \Pi_m = |w_m^*| \frac{\|\mathbf{G}_{:,m}\|_2}{\|\mathbf{b}\|_2}$$

Strong-identification criteria: continuity backbone in $[-1.01, -0.95]$; plausible signs; backbone stable across 6 scale triplets; $R^2_{\mathrm{wf},\rho} > 0.99$.

strong identification $\not\Rightarrow$ adequate closure

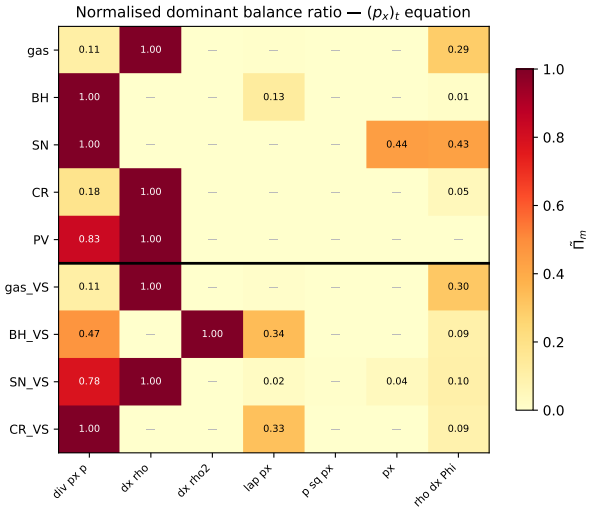
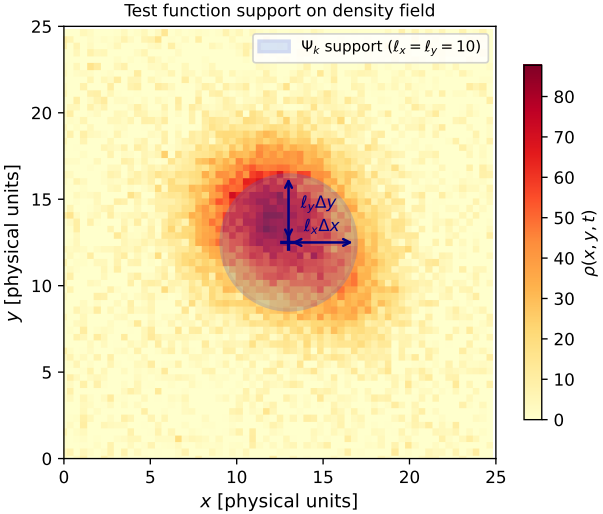
WSINDy Headline: Forecast Quality vs Identification Quality



Dominant Balance: Term Importance Hierarchy

Gas momentum: `rho_dx_Phi` $\tilde{\Pi} = 1.0$, `dx_rho` 0.85, `div_px_p` 0.42 (Morse dominates).
Pure Vicsek: `div_px_p` $\tilde{\Pi} = 1.0$, `dx_rho` 0.96 (no Morse term — exactly correct). Morse forcing dominates in attraction regimes; pure Vicsek is purely pressure-and-self-advection.

Momentum Equation: Cross-Regime Term Selection



Test-Function Footprint Sets the Effective Model Scale